



Diagnosing and Treating Malaria

With increased travel, please remember to consider malaria in the differential diagnosis in patients with prolonged fever and flu-like symptoms.

In addition to considering malaria in febrile patients with a history of travel to areas where malaria is endemic, a malaria diagnosis should now be considered in any person with a fever of unknown origin, regardless of travel history. Returning infected international travelers can transmit the parasite to domestic mosquitos when they get bitten, and these infected mosquitos, in turn, can pass the infection those who have not travelled.

So far, Florida and Texas confirmed five cases of domestic malaria caused by Plasmodium vivax, and it is just the beginning of summer and mosquito season. These are the first domestic cases since 2003.

Malaria Fast Facts

Malaria is a mosquito-borne illness transmitted through bites of infected female Anopheles mosquitoes.

Incubation Period

It’s usually 10 to 15 days after the bite, but it can be up to 4 weeks after infection.

Symptoms

- Fever, chills, and other flu-like symptoms (malaise, fatigue, diaphoresis, headache, cough, anorexia, abdominal pain, nausea, vomiting, diarrhea, arthralgias, and myalgias)
- Paroxysms of fever
- Fever may be higher than 104°F
- Patient may be tired, lethargic, jaundiced, or pale from anemia

P. vivax and P. ovale can have a dormant liver stage, resulting in activation weeks, months, or even years after initial infection. In these instances, mosquito bites during relapses can result in continued community spread of malaria.

If malaria is suspected, the patient should be evaluated in person urgently. Some species are fatal (P. falciparum and P. knowlesi).

Diagnostic Testing

Rapid antigen and blood smears

Name	Formulary	Sig/Dose	Code
PNL MALARIA AND BLOOD PARASITES AMB SCAL (aka MALARIA)			
PNL MALARIA AND BLOOD PARASITES AMB SCAL PLASMODIUM FALCIPARUM/PAN-MALARIA AG, RAPID TEST P Normal, Routine, BLOOD BLOOD PARASITES, SMEAR P Normal, Routine, BLOOD			

The process for ordering rapid antigen and blood smears to diagnose malaria.

Treatment

It’s complicated! We must consider travel history, drug resistance, the clinical status of the patient, and whether the patient got infected while on chemoprophylaxis for malaria.

Consult ID On Call and refer to the CDC website (the CDC treatment of malaria: Guidelines for clinicians may be particularly helpful). Confirmed cases are reportable.

References

National Institute of Allergy and Infectious Diseases. (2016). Life Cycle of the Malaria Parasite [Infographic]. National Institute of Allergy and Infectious Diseases. <https://www.niaid.nih.gov/diseases-conditions/malaria-parasite>

